Fanuc – Course Project Part 1 Documentation

Robot: FANUC LR Mate 200iD:

LINK DIMENSIONS

Base to J2 height :	330mm
J1 to J2 horizontal offset:	50mm
J2 to J3	330mm
J3 to forearm	35mm
J3 to wrist center	335mm
Wrist center to flange	80mm

ZERO POSE (All joints are = 0)

Upper arm points straight up

Forearm points straight forward

Flange x = 50 + 335 + 80 =465mm

Flange y = 0mm

Flange z = 330 +330 + 35 = 695mm

BASE FRAME

Base = base_link

Origin is J1 axis = 330mm below J2

Z is up

X is forward

FANUC’s own world frame 330mm higher, at j2 height

Z in FANUC’s world frame = z in base_link minus 330mm

FLANGE FRAME

The flange frame sit on link 6 without offset or rotation

Tool 0 is rotated = (180, -90, 0)Roll Pitch Yaw in Degs

At zero = tool z axis points forward(out of flange)

At zero = tool x axis is up

At zero = tool y axis points to -y

JOINT LIMITS AND SPEED

- J1: -170 to 170 deg, max 450 deg/s.
- J2: -100 to 145 deg, max 380 deg/s.
- J3: -70 to 205 deg, max 520 deg/s.
- J4: -190 to 190 deg, max 550 deg/s.
- J5: -125 to 125 deg, max 545 deg/s.
- J6: -360 to 360 deg, max 1000 deg/s.

FANUC TOT RANGE PER JOINT =

340, 245, 420, 380, 250, 720 deg

J3_kinematic= J2 + J3_pendant

J3_pendant is measured from base

ROS-Industrial, lrmate200id_macro.xacro (commit d8f42bd, 2025-02-19)
https://github.com/ros-industrial/fanuc/blob/noetic-devel/fanuc_lrmate200id_support/urdf/lrmate200id_macro.xacro

ROS wiki, fanuc_lrmate200id_support package page
https://whiteoak.umd.edu/roswiki/fanuc_lrmate200id_support.html

FANUC America, LR Mate 200iD product page
https://www.fanucamerica.com/products/robots/series/lr-mate/lr-mate-200id

mattj23 on GitHub, lrmate-200id (test_cases folder) https://github.com/mattj23/lrmate-200id

Robot-Forum, "How to get the calibrated Fanuc DH parameters from Fanuc Teachpendant"
https://www.robot-forum.com/robotforum/thread/51369-how-to-get-the-calibrated-fanuc-dh-parameters-from-fanuc-teachpendant/